

Agentic AI in CostVision

A costing tool that learns from every analysis, remembers your parts, and finds savings on its own — while staying fully auditable.

Remembers

Recognises

Self-corrects

Acts autonomously

EXECUTIVE SUMMARY

What we built — in one slide

36×

Error reduction

Estimating error fell from 10.9% to 0.3% after the tool learned from just 3 real quotes (verified live).

£512k/yr

Found autonomously

In our live demo the background agent flagged £512k/yr of pricing issues — with nobody at the keyboard.

99%

Part recognition

A new bracket was matched to 3 past bracket analyses at 98–99% similarity, with reasons shown.

790

Automated tests

Every capability is covered by automated tests and was exercised end-to-end on the running system.

The idea, in plain words

Until now, every costing started from zero and the result depended on who did it. **Now the tool keeps an organisational memory:** every analysis is stored, every real supplier quote teaches it, and every new part is compared against everything we have costed before.

The knowledge stays in our database, on our servers — it becomes a company asset that gets more valuable with use, and it does not walk out of the door when an expert leaves.

THE CONCEPT

What "Agentic AI" means here — four plain words



Remembers

Every costing is saved as a "case": the part, the inputs, the result, and any real quote. Shared across the whole team.



Recognises

Start a new part and it instantly finds the most similar past parts — like an experienced engineer saying "we've done this before".



Self-corrects

Log the real supplier price and the tool measures its own error, then corrects future estimates in that category. Accuracy is measured, not claimed.



Acts

A background agent re-checks all stored parts on a schedule and raises findings by itself: "you are overpaying here — worth £400k/yr."

Deliberate design choice: the learning is statistics over our own data — every suggestion shows its source parts and its arithmetic. That makes it auditable and defensible in front of a supplier, which is where costing tools win or lose.

HOW IT WORKS

How it works — the learning loop



...and every loop makes the next estimate better

No extra work for the engineer. Steps 2, 3, 4 and 6 happen automatically. The only new habit is one click — "Log Actual £" — when a real supplier quote arrives. That single click is the fuel for everything on this slide.

CAPABILITY 1 OF 5

The memory — an organisational knowledge base

What is stored for every analysis

- Part "fingerprint"** — process, material, weight, size, region, volume
- The full cost result** — total + breakdown by cost driver
- Real quotes** — actual supplier / PO prices, when logged
- Expert corrections** — where a person adjusted the AI's values
- CAD shape data** — dimensions & features, when a CAD file was used

Why it matters to us

✓ **Shared, not personal**

One engineer's analysis instantly helps everyone — juniors inherit senior judgement.

✓ **Stays on our servers**

Our database, our infrastructure. The knowledge is a company asset, not a vendor's.

✓ **No duplicates**

Re-costing a part updates its case — the memory stays clean.

✓ **Compounds with use**

Useful from ~20-30 analyses; every costing from today is an investment.

CAPABILITY 2 OF 5

Recognition — "we have costed this before"

Live example: an engineer costs a new 0.85 kg aluminium bracket. The tool answers instantly:

 **AI memory — similar past parts** (knowledge base: 3 analyses · 2 with actuals)

EV Battery Bracket 99% match — material family, weight, region **£41.20** **actual £46.50**

Sensor Mount Bracket 98% match — material family, weight, region **£38.50**

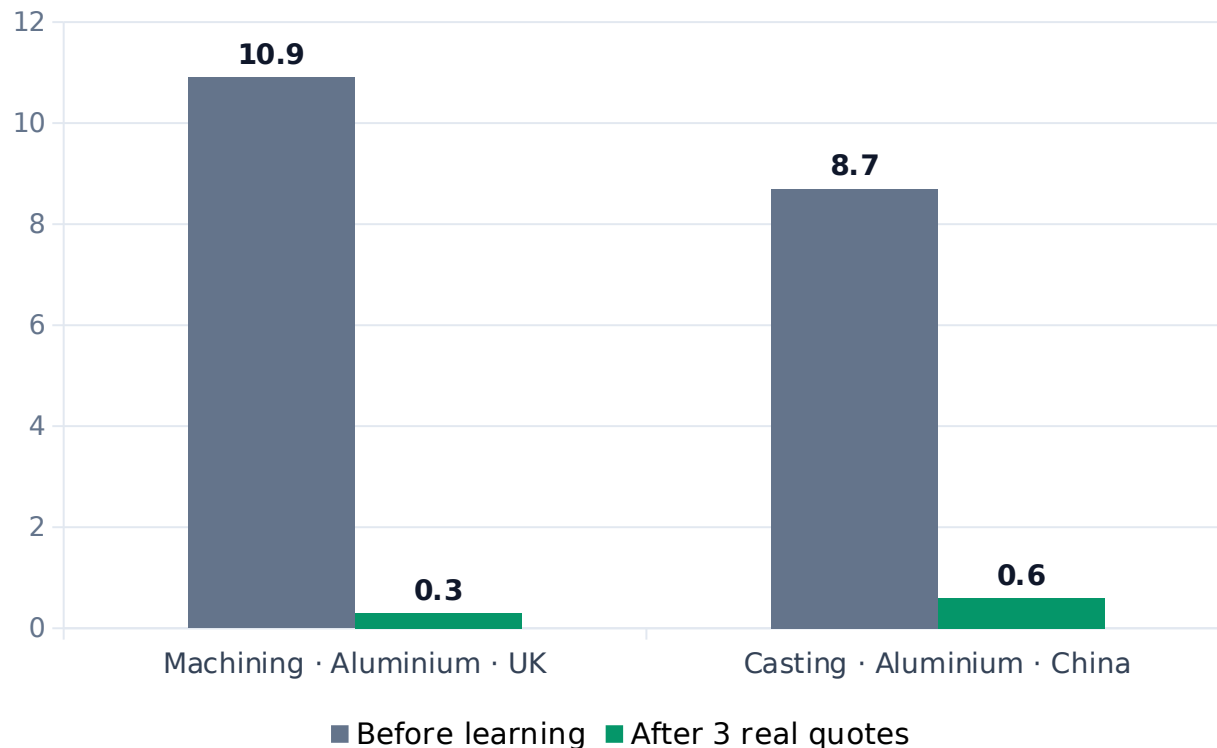
Inverter Bracket 98% match — material family, weight, region **£44.80** **actual £50.20**

- Median cost of 3 similar parts: £41.20
- 3 of 3 used the same material (aluminium 6061)
- Real quotes logged for 2 of them — median actual £48.35

Every match is explained (what matched, and how strongly) and every suggestion names its source parts — **no black box.**

CAPABILITY 3 OF 5

Self-correction — it learns from real quotes



Estimating error (%) — measured against real supplier prices, live system run

How it works, simply

1. A real supplier quote arrives → one click logs it.
2. The tool compares its estimate with reality and measures its own error.
3. From 3 quotes in a category, it corrects future estimates automatically — per process, material family and region.
4. It reports its accuracy openly (error % before and after).

Why per-category matters: our machining ran 12% low while our China castings ran 8% high — averaged together they looked "fine". The tool catches what averages hide.

CAPABILITY 4 OF 5

Honest ranges — from " $\pm 20\%$ " to " $\pm 3\%$ ", earned with data

Confidence band ($\pm \%$)



Width of the cost confidence band on the same part ($\pm \%$ around the estimate)

Every estimate now comes as a range

Optimistic (P10) · Most likely (P50) · Conservative (P90)

A single number implies a precision that early estimates don't have. The range tells buyers how much to trust the number — and what target to set in negotiation.

The band is earned, not guessed: it starts wide when the tool has no evidence, and tightens automatically as real quotes prove the accuracy. On our test part it went from $\pm 20\%$ to $\pm 3\%$ after three quotes.

CAPABILITY 5 OF 5

The autonomous agent — finds money unattended

A background monitor re-checks every stored part on a schedule. In the live demonstration it opened these findings **entirely unattended**:



RENEGOTIATION

≈ £400,000 / yr

Supplier price £48.00 is 20% above should-cost £40.00 at 50,000 pcs — clear recovery opportunity.



UNDERWATER PRICE

≈ £112,000 / yr exposure

Supplier price 25% BELOW should-cost — verify scope and quality; pricing may be unsustainable.



STALE ESTIMATE

Confidence decay

A 120-day-old estimate was never validated with a real quote — the agent asks for a refresh.

Total surfaced in the demo: ≈ **£512,000 / yr** — every finding shows its arithmetic, and one click dismisses it once handled.

THE WIDER AI PLATFORM

Supporting AI capabilities already in the tool



Grounded AI assistant

Answers cite our actual rate library and past costings — the AI quotes our data, it doesn't improvise. Every figure is traceable.



RFQ analyst

Drop in an RFQ package: it costs every line, flags risky prices and single-source parts, and drafts a prioritised negotiation brief.



CAD feature costing

Reads the 3D model and prices individual design features — holes, threads, surfaces — so designers see exactly what drives cost.



Carbon co-costing

Every part gets a CO₂e figure alongside the £ — increasingly demanded in automotive & aerospace RFQs.

Together with 18 commodity cost engines, CAD-to-cost, and PCB photo-to-BOM — the agentic layer sits on top of all of it.

INPUTS REQUIRED

What it needs from us — honestly, very little

You provide

- 1. Keep costing parts as normal**
every analysis feeds the memory automatically
- 2. One click when a real quote arrives**
"Log Actual £" — this is the learning fuel
- 3. Optional: CAD file or BOM**
sharper matching and better auto-fill
- 4. Optional: historic quotes (CSV)**
seeds the memory so it starts smart, not empty

The tool does

- ✓ Remembers every analysis (no duplicates, org-wide)
- ✓ Finds & explains similar past parts instantly
- ✓ Suggests benchmarks, materials and real prices
- ✓ Measures its own error and self-corrects per category
- ✓ Tightens confidence ranges as evidence grows
- ✓ Monitors all parts in the background & raises findings

EVIDENCE

Results & accuracy — measured, not promised

Estimating error after learning (machining segment)

10.9% → 0.3%

Estimating error after learning (casting · China)

8.7% → 0.6%

Confidence band on the same part

±20.4% → ±2.8%

Similar-part recognition on live example

98-99% match, reasons shown

Autonomous findings in unattended demo

£512,000 / yr surfaced

Automated tests protecting all of this

790 passing (60 suites)

How we verified: every number above comes from running the real system end-to-end — live server, real database, real API calls — not from slides or simulations.

BENEFITS

What this means for the business

⚡ Faster costing

New parts start from proven history instead of a blank sheet — matches, materials and benchmarks appear instantly.

🎯 Accuracy that compounds

Every real quote makes the next estimate better. Accuracy is measured and reported — defensible in any negotiation.

🏛️ Knowledge stays

Senior engineers' judgement is captured as data. It doesn't leave when people do — and juniors inherit it from day one.

💰 Money found proactively

The agent watches all parts continuously and flags overpayment and pricing risk by itself, quantified in £/yr.

And it is ours: the knowledge base runs on our infrastructure and grows into a proprietary asset competitors cannot buy.

NEXT STEPS

Where we are, and the ask

Status today

- ✓ All 6 capabilities built, tested (790 tests) and live
- ✓ Verified end-to-end on the running system
- ✓ Zero extra licence cost — built into our tool
- ✓ Runs on-premise; no data leaves the company

Honest note: the intelligence starts empty and grows with use. The mechanism is proven; the value builds as we feed it.

The ask — three small decisions

1. Adopt the habit

Make "Log Actual £" part of the quote-handling routine. One click per quote.

2. Seed the memory

Approve a one-off import of historical quotes so the tool starts smart (~1-2 days of effort).

3. Review the findings

Put the agent's monthly findings on the sourcing team's agenda — it is already finding money.